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## Dataset Acquisition



## LVIS

1000 images ·  
4,132 GT image-  
category pairs

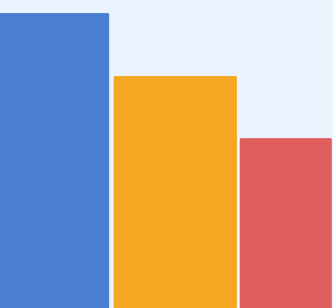


## COCO

1000 images ·  
3,217 GT image-  
category pairs



Long-tail/frequency analysis



Rare Common Frequent

Dataset evaluation  
subsets

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## Data Preprocessing



Image Resize  
& Normalise

OWL-ViT image processor



Annotation  
Parsing

Ground-truth annotations  
used for evaluation



Evaluation protocol setup



Frequency-  
aware prior only  
for LVIS

Semantic Fusion Framework

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## Semantic Prompt Generation



WordNet  
prompts

WordNet semantic variants  
and related terms  
semantic variants



Template  
prompts

attributes · contexts  
descriptions

airplane

visible  
airplane

aeroplane

vehicle

aircraft

photo of  
airplane

prompt variants per  
category

Complementary + redundant  
semantic sources

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## LVIS frequency-prior score recalibration



**Image Stream**  
OWL-ViT visual  
encoder  
Visual features  
for detection

image evidence



**Text Stream**  
OWL-ViT text  
encoder  
Prompt  
embeddings  
per class  
Image-text  
matching



## Semantic Fusion Strategies

Mean

Max

Weighted

Adaptive

Entropy/conflict risk estimation  
Frequency-aware LVIS reweighting

Redundancy-complementarity analysis  
(RQ3-RQ4)



Image-level and pair-level semantic evidence scoring  
Matched against GT categories and IoU AP50-proxy  
where available

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## Experimental Evaluation



Evaluation  
Metrics

Image-level recall · pair recall ·  
AP50-proxy



Ablation Studies

Fusion strategy comparison  
Prompt-source removal  
Prior-aware effect



Framework evolution analysis  
Primary → semantic → adaptive →  
prior-aware

Example semantic scores

airplane 0.90

aircraft 0.84

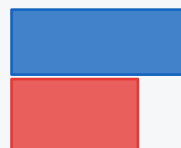
vehicle 0.61

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## Result &amp; Impact

LVIS semantic evidence  
recall: 0.9131

Semantic evidence vs.  
primary-prompt baseline



Semantic  
evidence

Primary  
prompt

Recall/AP50-proxy  
comparison

Uncertainty-risk bins



Findings support safe,  
bounded claims